

# Reinforcement Learning for Option Pricing

Kristian Boroz

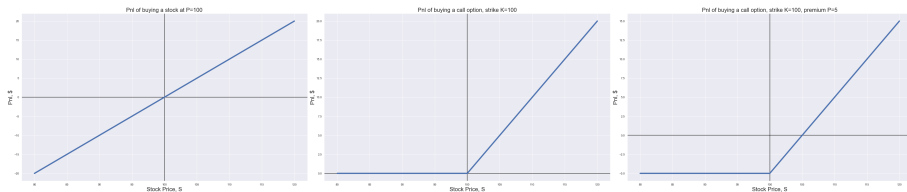
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Sommer-Semester 2023

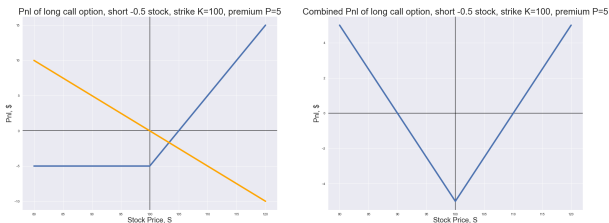
# Outline

- 1 Motivation
  - Models
  - Calibration
- 2 Reinforcement Learning
  - Main Ideas
  - Implementation
  - Results

# From Options and Calibration To Hedging



**Abbildung:** Stockprice and the PayOff-Profile for a Call-Option



**Abbildung:** Profit-Loss of Combined Position

# Final Goal: Improving the Hedging-Performance

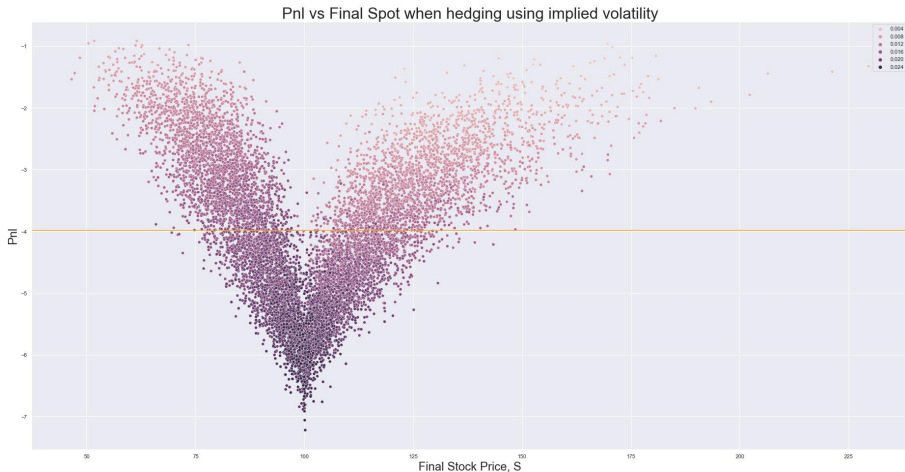


Abbildung: Profit-Loss of Hedging Position

# Black-Scholes-Dynamics - Stockmarkets

$$dS_t = \mu S dt + \sigma S dW \quad (1)$$

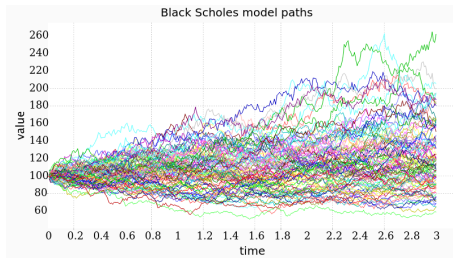


Abbildung: Black-Scholes sample paths

# Vasicek-Dynamics - Interest Rate Products

$$dr_t = \alpha(\beta - r_t)dt + \sigma dW \quad (2)$$

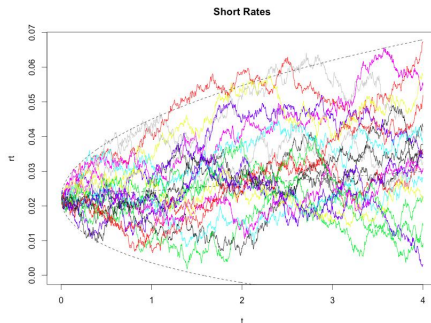


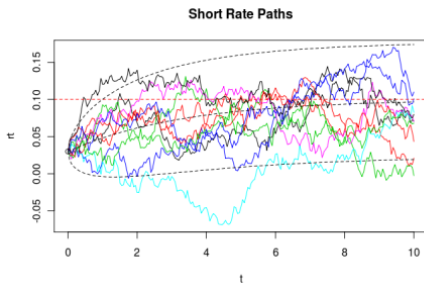
Abbildung: Vasicek sample paths

# Heston Dynamics - Advanced Model

$$dS_t = \mu S_t dt + \sqrt{V_t} S_t dW_t^S \quad (3)$$

$$dV_t = \kappa(\theta - V_t)dt + \xi \sqrt{V_t} dW_t^V \quad (4)$$

$$dW_t^S W_t^V = \rho dt \quad (5)$$



# Real Market Data - SP500

## S&P 500 (^GSPC)

SNP - SNP Real Time Price. Currency in USD

☆ Follow

**4,455.59** +5.21 (+0.12%)

At close: July 3 02:23PM EDT

Start Trading >>

Plus500 86% of retail CFD accounts lose money

Summary

Chart

Conversations

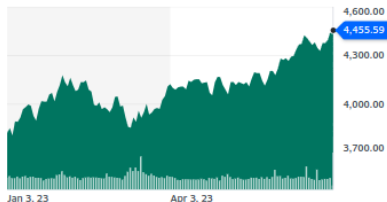
Historical Data

Options

Components

|                |               |               |                     |
|----------------|---------------|---------------|---------------------|
| Previous Close | 4,450.38      | Day's Range   | 4,442.29 - 4,456.46 |
| Open           | 4,450.48      | 52 Week Range | 3,491.58 - 4,458.48 |
| Volume         | 1,243,004,000 | Avg. Volume   | 3,988,518,333       |

1D 5D 1M 6M YTD 1Y 5Y Max Full screen





# Real Market Data - VIX

## CBOE Volatility Index (^VIX)

Cboe Indices - Cboe Indices Delayed Price. Currency in USD

☆ Follow

**13.57** -0.02 (-0.15%)

At close: July 3 12:15PM CDT

Advertisement

Start Trading >>

Plus500 86% of retail CFD accounts lose money

Summary

Chart

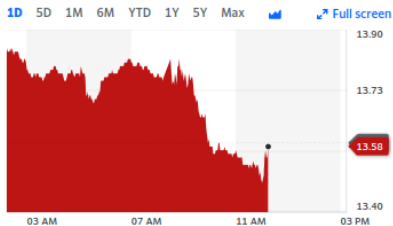
Conversations

Historical Data

Options

Components

|                |       |               |             |
|----------------|-------|---------------|-------------|
| Previous Close | 13.59 | Day's Range   | 0.00 - 0.00 |
| Open           | 0.00  | 52 Week Range |             |
| Volume         | 0     | Avg. Volume   | 0           |



# Real Market Data - Reality

## SPDR S&P 500 ETF Trust (SPY)

NYSEArca - Nasdaq Real Time Price. Currency in USD

☆ Follow

**443.79** **+0.51 (+0.12%)** **443.95** **+0.16 (+0.04%)**

At close: July 3 01:00PM EDT

After hours: Jul 3, 04:59PM EDT

[Summary](#) [Chart](#) [Conversations](#) [Historical Data](#) [Profile](#) [Options](#) [Holdings](#) [Performance](#) [Risk](#)


**Hannover** **OPEN** | 6AM-10PM  
Borner Straße 66A, Hannover



July 5, 2023



In The Money

Show: **List**[Straddle](#)

Option Lookup



### Calls for July 5, 2023

| Contract Name      | Last Trade Date        | Strike ^ | Last Price | Bid   | Ask   | Change | % Change | Volume | Open Interest | Implied Volatility |
|--------------------|------------------------|----------|------------|-------|-------|--------|----------|--------|---------------|--------------------|
| SPY230705C00373000 | 2023-06-29 10:27AM EDT | 373.00   | 63.96      | 70.62 | 71.24 | 0.00   | -        | -      | 1             | 105.08%            |
| SPY230705C00390000 | 2023-06-29 11:55AM EDT | 390.00   | 48.05      | 53.62 | 54.24 | 0.00   | -        | -      | 0             | 81.05%             |
| SPY230705C00396000 | 2023-06-27 4:01PM EDT  | 396.00   | 40.30      | 47.62 | 48.24 | 0.00   | -        | -      | 0             | 72.75%             |
| SPY230705C00400000 | 2023-07-03 12:20PM EDT | 400.00   | 43.83      | 43.62 | 44.25 | -0.17  | -0.39%   | 5      | 4             | 67.58%             |
| SPY230705C00406000 | 2023-06-27 4:01PM EDT  | 406.00   | 30.44      | 37.63 | 38.25 | 0.00   | -        | -      | 0             | 59.47%             |
| SPY230705C00408000 | 2023-06-30 1:20PM EDT  | 408.00   | 35.63      | 35.63 | 36.25 | 0.00   | -        | 1      | 0             | 56.64%             |
| SPY230705C00410000 | 2023-07-03 1:11PM EDT  | 410.00   | 33.85      | 33.59 | 34.25 | -0.02  | -0.06%   | 184    | 0             | 52.64%             |

# Black-Scholes Formula - Call Price

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$$Call = C(S, t) = S\Phi(d_1) - Ke^{-r(T-t)}\Phi(d_2) \quad (6)$$

$$d_1 = \frac{\ln(S/K) + (r + \sigma^2/2)(T - t)}{\sigma\sqrt{T - t}} \quad (7)$$

$$d_2 = \frac{\ln(S/K) + (r - \sigma^2/2)(T - t)}{\sigma\sqrt{T - t}} \quad (8)$$

$$\Phi(x) = \int_{-\infty}^x \frac{1}{\sqrt{2\pi}} \exp\left(-\frac{z^2}{2}\right) dz \quad (9)$$

$$Delta = \Delta = \frac{dC}{dS} = \Phi(d_1); Vega = \frac{dC}{d\sigma} \quad (10)$$

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# Option Pricing - Revised

European Call Option: Right to buy on future data at agreed price

European Put Option: Right to sell on future data at agreed price

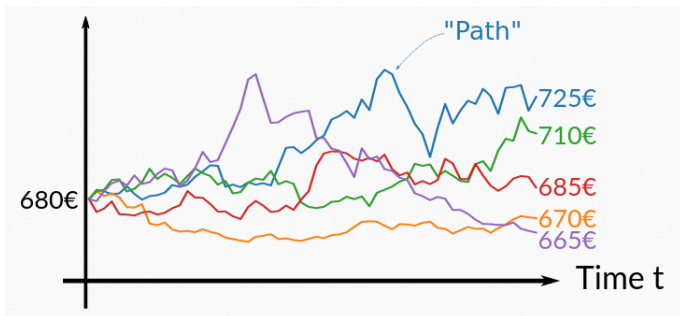
American Option: One can exercise an option on any day before expiration

Knock-In and Knock-Out:

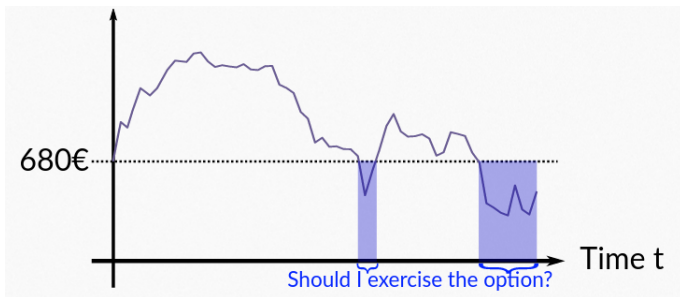
(In)-options only become active in the event that a predetermined knock-in barrier price is breached.

(Out)-options start their lives active and become null and void in the event that a certain knock-out barrier price is breached.

# Exercise Boundary for American Call

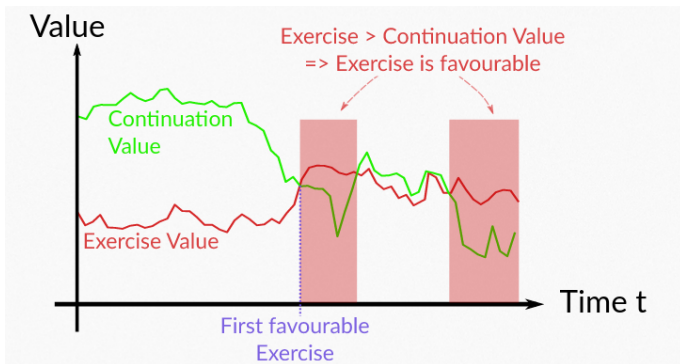


# Exercise Boundary for American Call

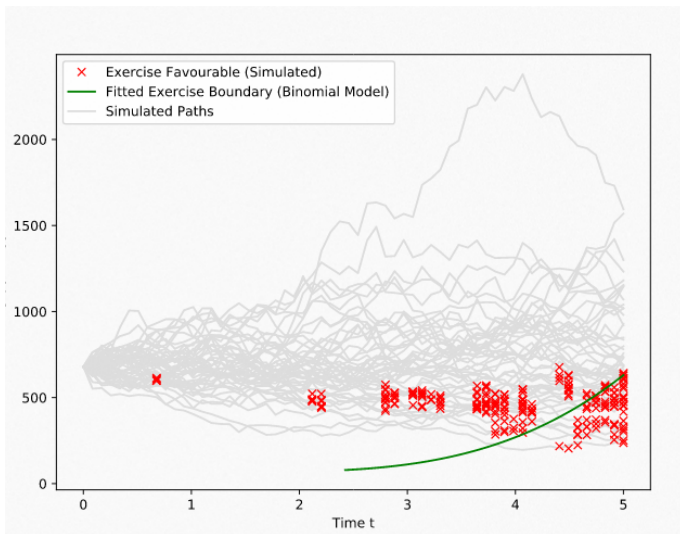




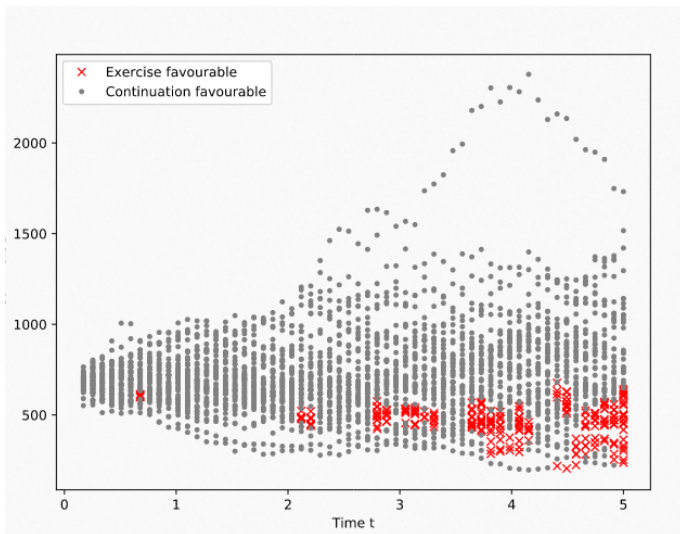
# Exercise Boundary for American Call



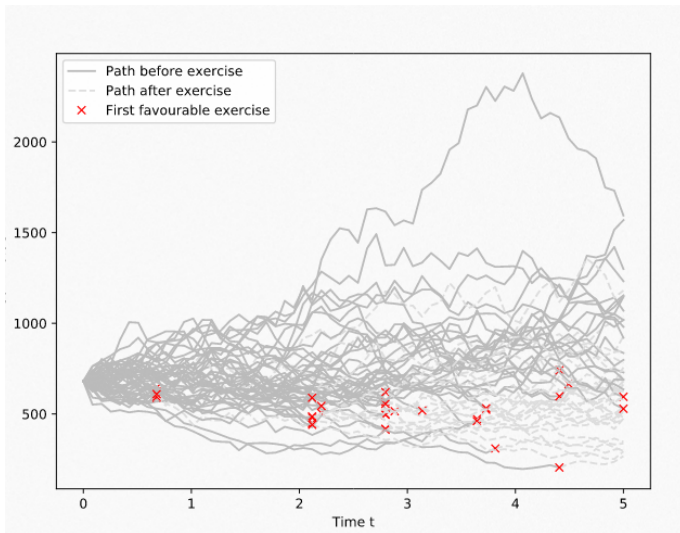
# Exercise Boundary for American Call



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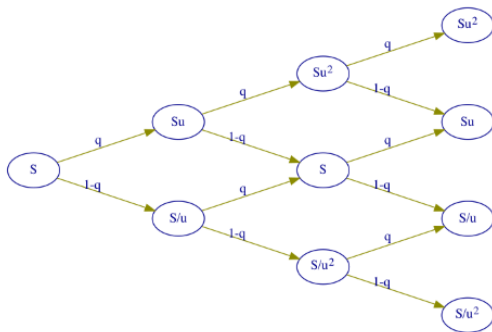


# Exercise Boundary for American Call



# Discretization of Grid - DPP - BS-Options

$$V(S_t, t) = \exp(-r\Delta t)P_u V(S_u, t + \Delta t)P_d V(S_d, t + \Delta t)$$



Black Scholes price: 0.10475067856063891

Simulated price: 0.10303646

# Discretization of Grid - DPP - US-Knock-In-Options

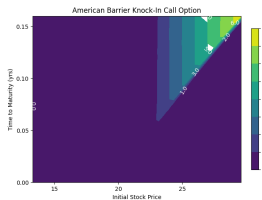
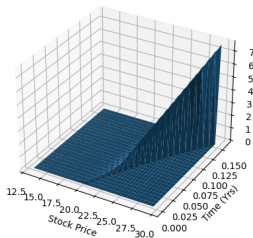
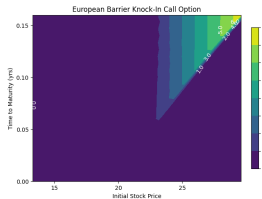
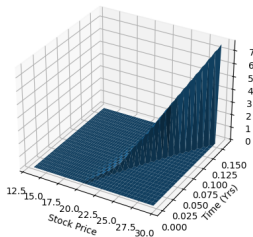
$$Q(S, t, a_t = \text{exercise}) = P_h(S, t) \max(S - K, 0)$$

$$Q(S, t, a_t = \text{hold})$$

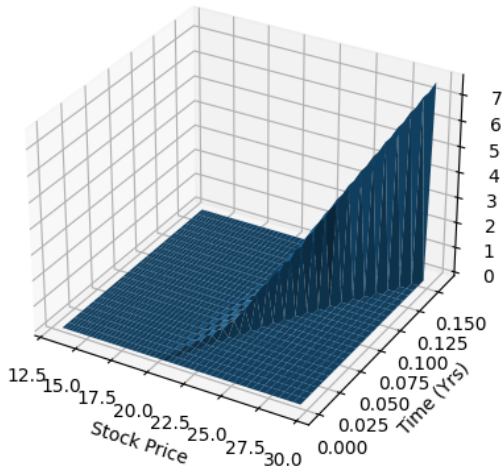
$$\exp(-r\Delta t) P_h(S_t, t) \left( P_u \frac{V(S_u, t + \Delta t)}{P_h(S_u, t + \Delta t)} + P_d \frac{V(S_d, t + \Delta t)}{P_h(S_d, t + \Delta t)} \right)$$

$$V(S_t, t) = \max(Q(S, t, a_t = \text{exercise}), Q(S, t, a_t = \text{hold}))$$

# Knock-In-Options - EU vs. US

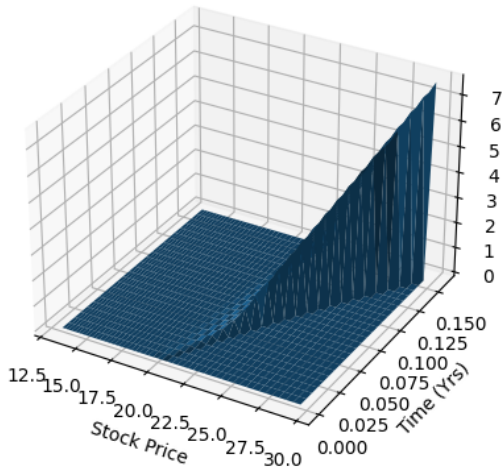


# Knock-In-Options - EU vs. US

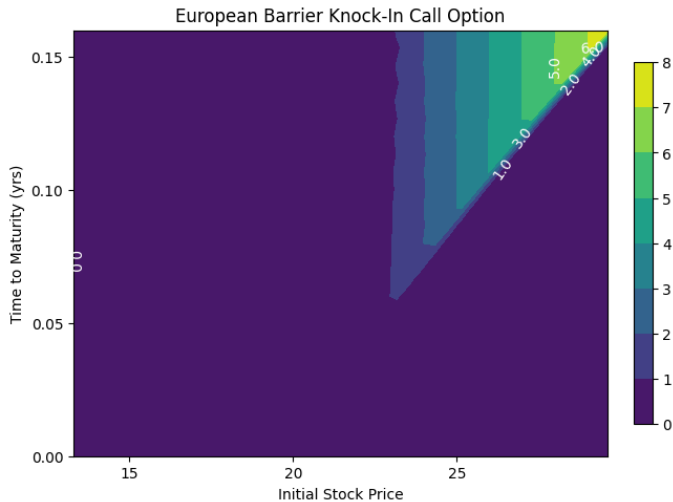




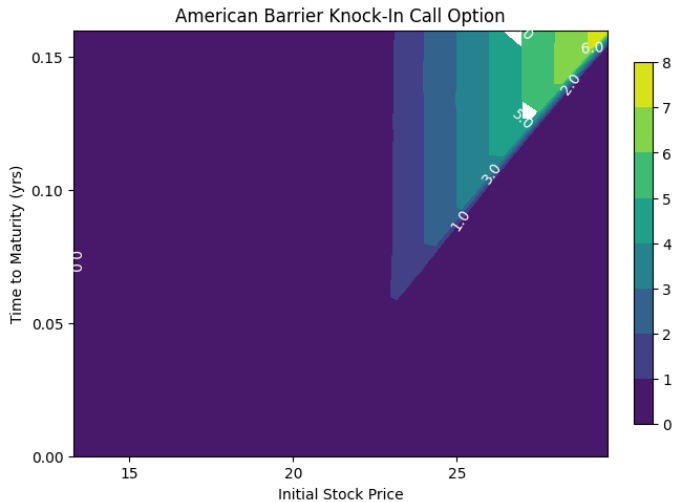
# Knock-In-Options - EU vs. US



# Knock-In-Options - EU vs. US



# Knock-In-Options - EU vs. US



# Summary

- Reinforcement Learning enables the pricing of Options with a non-linear Payoff-profile
- Computation-times are OK
- Manifold extensions are possible

# Literaturverzeichnis

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